DSA03-Natural Language Processing

List of Lab Program

15. Implement probabilistic context-free grammar parsing for a sentence using python.

Code:

```
import nltk
from nltk import PCFG
from nltk.parse import ViterbiParser
grammar = PCFG.fromstring("""
S -> NP VP [1.0]
NP -> Det N [1.0]
VP -> V NP [1.0]
Det -> 'a' [0.6] | 'the' [0.4]
N -> 'cat' [0.5] | 'mouse' [0.5]
V -> 'chased' [0.7] | 'caught' [0.3]
""")
parser = ViterbiParser(grammar)
sentence = ['the', 'cat', 'chased', 'a', 'mouse']
print("Sentence:", " ".join(sentence))
print("Parse Tree:")
for tree in parser.parse(sentence):
    print(tree)
    print("Probability:", tree.prob())
```

Output:

```
Sentence: the cat chased a mouse
Parse Tree:
(S
  (NP (Det the) (N cat))
  (VP (V chased) (NP (Det a) (N mouse)))) (p=0.042)
Probability: 0.042
```